CS107.3 – Object Oriented Programming with C#

Faculty of Computing

NSBM GREEN UNIVERSITY TOWN

ASSESMENT GUIDELINES

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Overview

This document contains all the necessary information pertaining to the assessment of ***CS107.3 Object Oriented Programming with C#.*** The module is assessed via **50% Examination and 50% Coursework**.

The sections that follow will detail the assessment tasks that are to be undertaken. The submission and expected feedback dates are presented in Table 1. All assessments are to be submitted electronically via the respective nLearn module pages before the stated deadlines.

	Submission Deadline	Feedback
Coursework Proposal	**2nd Week of June 2026**	During Lectures / As groups in predetermined time slots.
Report + Coursework +Video	**9th of July 2026**	During Viva

Table 1: Assessment Deadlines

All assessments will be introduced in class to provide further clarity over what is expected and how you can access support and formative feedback prior to submission. Whilst the assessment information is provided at the start of the module, it is not necessarily expected you will start this immediately – as you will often not have sufficient understanding of the topic. The module leader will provide guidance in this respect.

Assessment 1: Coursework

Task:

This assignment contributes **50%** of the overall module mark for CS 107.3 – Object Oriented Programming with C# and is a **Group assignment**. The **final submission** must be submitted to nLearn by the specified submission date.

The groups should consist of **8 – 10 members** per group. Each member is expected to contribute to the development of the application and should be aware of the solution in general.
The groups should be finalized and registered in nLearn before **3rd of June** 2026.

This is a negotiated project in which you must define the specific content you will produce and the process you will follow, within the guidelines given below. Your project **must satisfy** the following requirements:

- Comprehensive coverage of OOP concepts.
- Exception handling.
- Database Integration

You must produce **a working system**, comprising a client application. The application must be developed using C# as the main development language and target the .NET framework. The system must be interactive, i.e., the user should be able to affect its behavior by interacting with it using a keyboard, mouse, or another suitable input device. The final product should reflect an effort of **40+ hours of dedicated work** (including the usage of version controlling platforms).

You must document the system, including its design and details of the implementation process you have followed, and provide a description of your version controlling techniques used. This could include your code repository, snapshots of your database, etc.

To differentiate each project and foster innovation, **each system must incorporate at least one unique, innovative feature that sets it apart from conventional existing solutions.** This feature should address a specific gap, enhance user experience significantly, or introduce a novel functionality relevant to the chosen domain.

You must choose ***one*** of the following application topics for your project:

1. **Ayurveda Clinic Management System**: Specialized for traditional Sri Lankan Ayurvedic practices, managing patient profiles, herbal prescriptions, and therapy schedules.
2. **Temple Event & Resource Management System**: A system designed for Buddhist temples in Sri Lanka to manage religious ceremonies, dana offerings, monastic requisites, and visitor schedules.
3. **Tea Estate Crop & Labour Management System:** Tailored for Sri Lanka's tea industry, focusing on tea plucking cycles, yield tracking per estate section, worker attendance, and green leaf processing.

4. **Coastal Fishery Fleet & Catch Tracker:** A system for Sri Lankan fishing communities to manage boat movements, track daily catch data, monitor fish stock, and predict weather patterns relevant to fishing.
5. **Gem Mine Operations & Certification System:** Specifically for Sri Lanka's gem industry, managing mine plot allocations, extracted raw gem tracking, cutting and polishing schedules, and certification for authenticity.
6. **Paddy Field Yield & Irrigation Planner:** A system specifically designed for Sri Lankan paddy farmers to manage crop cycles, predict yields based on weather patterns, and optimize irrigation schedules for rice cultivation.
7. **Cultural Heritage Site Management System:** For managing and preserving Sri Lanka's historical and archaeological sites, including visitor tracking, artifact inventory, conservation project oversight, and tour guide scheduling.
8. **Spice Plantation Management System:** Focused on the unique requirements of Sri Lankan spice cultivation (e.g., cinnamon, pepper, cardamom), tracking planting, harvesting, drying processes, quality control, and export logistics.
9. **Local Artisan & Craft Market Platform:** An e-commerce and management system supporting Sri Lankan local artisans, allowing them to showcase traditional crafts, manage orders, track material sourcing, and facilitate direct sales.
10. **Tuk-Tuk Driver Dispatch System:** A localized solution for Sri Lanka's ubiquitous three-wheeler (tuk-tuk) services, including driver registration, driver allocation, vehicle traffic prediction, and customer booking.

COURSEWORK DELIVERABLES

There are two main deliverables for this assignment. Only deliverable **D1**, the final submission, and **D2**, the Demonstration and Viva, will contribute to the final module mark.

D1 – Final Submission
For this deliverable, you must submit a ZIP archive containing your source code.
There is a 50MB limit on the final submission. Please check your submitted files are correct by downloading them again and ensuring they work. You will receive a confirmation receipt by email when your work has been properly submitted – if you do not receive this email, your work has not been submitted.
The ZIP archive should contain all of the code you have written yourself and should, as far as possible, contain copies of the folders and files needed to run your application.

You must include the following elements in the final submission:

1. Source Code:
This should contain all of the code you have written yourself and should, as far as possible, contain copies of the folders and files needed to run your application.

2. Report:
The report must be a document of no more than 2,000 words. Please use screen shots and sketches to illustrate the functionality and UML diagrams (or appropriate alternatives) to illustrate the design. The report should explain:

- <u>Problem statement (ca. 400 words with additional documents in appendices)</u>
  - Existing problem that you wish to address.
  - How will this contribute as a solution.
- <u>Requirements (ca. 400 words with additional documents in appendices)</u>
  - Who is the application aimed at?
  - What features were included and why?
- <u>Design (ca. 400 words with UML diagrams)</u>
  - What is the system architecture.
  - How do the processes interact?
  - How are the data and code structured?
  - Why are these structures appropriate?
- <u>Personal reflection (ca. 400 words)</u>
  - What worked/didn't work well in terms of your work and the technologies you used?
  - What lessons would you take from this project into your next project.
- <u>Future work (ca. 400 words)</u>
  - How could this application be developed further.
  - What changes will you make in order to make it a more sustainable product.

D2 – Demonstration and Viva

You are required to create a video demonstration of your application. This video should:

- Clearly demonstrate the functionality of your application.
- Explain the implementation process, including both the development process and the application functionality.
- Be uploaded to YouTube as an *unlisted* video (duration: **5 min**).
- The link to this unlisted YouTube video must be included in your report.
- The video must also be submitted separately via nLearn.

This deliverable also includes a viva (oral examination), where you will be questioned about your project.

Assessment Criteria:

Your work will be assessed according to the rubric found in Table 1. Your mark for this piece of coursework will be based on an aggregation of the marks for each category. Marks will be awarded based on **both the demonstration and the report**.

NOTE: ***Each Member of the Group Should have sufficient Understanding of the whole solution. Including all the parts & integration.***

Category	**Fail (< 40%)**	**>= 40%**	**>= 50%**	**>= 60%**	**>= 70%**
Analysis (20%)	Insufficient problem analysis. Functional requirements are not described in sufficient detail.	There are some vague functional requirements based on a basic outline of the problem.	Reasonably detailed functional requirements but based on weak analysis.	Functional requirements are well specified with some evidence of strong analysis.	Functional requirements are complete and are well motivated based on strong analysis.
Design (25%)	Little or no design work. It is not clear how the system will be built, or what its major components are.	Some overview of the design but no detail. The architecture is not defined. Little or no use of diagrams to show the structure and operation of specific sections of the system.	Some of the system's design is specified in detail but other areas are weak. Some diagrams are included.	Design is mostly complete, further detail would enhance some areas. Most functionality and design specified with diagrams.	System design complete and comprehensively described with the use of appropriate diagrams.
Development (30%)	Very little software has been constructed, and what there is does not work.	Most of the functionality has not been implemented. Major errors at runtime. OOP concepts and/or a database have been attempted but incomplete. Much of the system is affected by bugs.	Some requirements implemented but major omissions. Software contains major runtime errors. OOP concepts and database used, one of them weakly the software has some bugs affecting the main functionality.	Largely complete implementation of most requirements. Some runtime errors. OOP concepts and a database are used, both works correctly. There are some apparent bugs, but they do not impede the main functionality.	Implementation of all requirements is complete. Very few apparent bugs. The approach taken to implementing the database and usage of OOP concepts is nearing or at a commercial standard.
Individual Contribution (15%)	Minimal or no contribution. Little evidence of individual work. Individual knowledge & continuous participation are insufficient.	Some contribution is evident but lacks clear differentiation from group work. Limited individual knowledge & sporadic participation.	Contribution is clear but only meets basic expectations. Demonstrates basic individual knowledge & fairly consistent participation.	Contribution is substantial, showing initiative and more than basic effort. Demonstrates good individual knowledge & regular participation.	Contribution is significant, clearly impactful, and demonstrates leadership and independence. Demonstrates strong individual knowledge & continuous participation.
Presentation Quality (10%)	Presentation is disorganized, unclear, and unengaging. Visual aids are poorly used or absent.	Presentation is basic with some clarity issues. Visual aids are minimal or not effectively used.	Presentation is clear and organized with reasonable use of visual aids.	Presentation is clear, well-organized, and engaging. Visual aids are effectively used.	Presentation is highly engaging, clear, and exceptionally well-organized. Visual aids are used creatively and effectively.

Table 1: Feedback Template for Assessment 1

General Guidance

Plagiarism

All of your work must be your own. You must use references for your sources; however, you acquire them. Where you wish to use quotations, these must be a very minor part of your overall work.

To copy another person's work is viewed as plagiarism and is not allowed. Any issues of plagiarism and any form of academic dishonesty are treated very seriously. All your work must be your own and other sources must be identified as being theirs, not yours. The copying of another persons' work could result in a penalty being invoked.

Submission Guidelines

Ensure that you follow all submission guidelines provided for your coursework. This includes file formats, word limits, and any specific instructions related to the submission process. Late or improperly formatted submissions may result in penalties.

Penalties for Plagiarism

Understand the potential consequences of plagiarism and academic dishonesty. Penalties can range from receiving a failing grade on the assignment to more severe academic sanctions, such as suspension or expulsion. It is vital to approach your academic work with integrity and honesty.

Final Checks

Before submitting your work, perform a final check to ensure that all sources are properly cited, your quotations are correctly attributed, and your work adheres to the guidelines provided. Reviewing your work carefully can help avoid accidental plagiarism and ensure that your submission meets academic standards.

Referencing

Proper referencing is crucial for academic integrity and helps to avoid plagiarism. Use a consistent citation style throughout your work. Common citation styles include APA, MLA, Chicago, and Harvard. Ensure that every source you use is accurately cited in both in-text citations and the reference list.

Use of Quotations

Quotations should only be used to support your argument and must be properly attributed to their original source. They should not dominate your work. Aim to paraphrase and summarize information in your own words where possible and use direct quotations only when the original wording is essential for understanding or emphasis.